

Yingheng Wang

CS PhD Candidate (AI/ML) | Cornell University

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Education

Cornell University

Aug. 2022 - present

Advisors: [Carla P. Gomes](#), [Volodymyr Kuleshov](#), [Christopher De Sa](#)

Ithaca, NY

Ph.D. in CS (AI & ML) | GPA: 3.91/4.0

M.S. earned en route | GPA: 4.0/4.0

Johns Hopkins University

Aug. 2021 - Jun. 2022

Baltimore, MD

M.S. in BME (dual degree earned en route) | GPA: 4.0/4.0

Tsinghua University

Sep. 2020 - Jun. 2022

Beijing, China

M.S. in EE (exam-exempt admission) | GPA: 4.0/4.0

Tongji University

Sep. 2016 - Jun. 2020

Shanghai, China

B.S. in ECE (president's award) | GPA: 95/100 (Rank: 1/110)

Research Interest

My research centers on developing knowledge-centric AI methods for scientific reasoning, which integrate domain scientific knowledge into modern AI techniques to improve their efficacy, efficiency, robustness, and interpretability. Current research topics include:

- **Learning with scientific knowledge:** learning with constraints, learning to discover.
- **Learning under limited resources:** self-supervised and unsupervised learning, multimodal learning, expressive and efficient neural architecture.
- **Reasoning and inference:** reinforcement learning, constrained reasoning, probabilistic inference.
- **Generative AI:** large language models, diffusion and flow-based generative modeling, AI agents.
- **Applications:** scientific discovery in chemistry, biology, materials, life science, and sustainability.
- **Taxonomy, Datasets, and Benchmarks:** taxonomy of generative molecule design, dataset and benchmark of machine learning for materials discovery, taxonomy of structured data augmentation, evaluating combinatorial LLM reasoning, evaluating spatial LLM reasoning in molecule discovery.

These thrusts converge to build AI systems that can reason, learn efficiently, and generalize robustly in complex real-world scenarios, ultimately advancing scientific discovery and innovation.

Research Impact

I have published research in leading AI/ML and science venues, including ICML, NeurIPS, ICLR, AAAI, TMLR, Nature, and Nature Machine Intelligence. My work has garnered 2,000+ citations, with several papers receiving Best Paper Awards, Featured Paper Awards, and winning ML Challenges. I was also student researcher at Amazon Grand Challenge, AWS AI Lab, Microsoft Research, and NEC Lab America, where my research led to impactful papers and patents on computational antibody design, protein language models, time-series foundation models, and efficient neural architectures.

Publications (*Equal contribution. [†]Correspondence.)

Selected:

[1] [Yingheng Wang](#)[†], Francesco Ricci[†], Zhilong Wang, Tao Yu, Junwen Bai, Xiangyun Lei, Shufeng Kong, Fengqi You, John M. Gregoire, Carla P. Gomes[†]. “DeepDOSReasoner: Physics-Grounded Reasoning for Den-

sity of States Prediction". [Nature Computational Science](#), 2026 [Under Review](#)

[2] **Yingheng Wang**, Can Chen, Karla-Luise Herpoldt, Sukanya Sasmal, Tommi Jaakkola, Marcus D. Collins, Shang Shang. "AbMPO: Multi-Objective Policy Optimization for Affinity-Calibrated Antibody Sequence Design". [NeurIPS](#), 2026 [Under Review](#)

[3] **Yingheng Wang**, Tao Yu, Shufeng Kong, Francesco Ricci, John M. Gregoire, Carla P. Gomes. "Deep Scientific Reasoning under Physical Constraints: Structure-Aware Spectrum Prediction". [ICML](#), 2026

[4] Hongzheng Chen*, **Yingheng Wang***, Yaohui Cai*, Hins Hu*, Jiajie Li*, Shirley Huang, Chenhui Deng, Rongjian Liang, Shufeng Kong, Haoxing Ren, Samitha Samaranayake, Carla P. Gomes, Zhiru Zhang. "HeuriGym: An Agentic Benchmark for LLM-Crafted Heuristics in Combinatorial Optimization". [ICLR](#), 2026 [\[paper\]](#) [\[code\]](#)

[5] **Yingheng Wang**, Zichen Wang, Gil Sadeh, Luca Zancato, Alessandro Achille, George Karypis, Huzefa Rangwala. "LC-PLM: Long-context Protein Language Modeling Using Bidirectional Mamba with Shared Projection Layers". [ICLR](#), 2026 [\[paper\]](#) [\[code\]](#) 🏆 [TMLR Featured Certificate](#)

[6] **Yingheng Wang***, Top Wasu Piriyaikulij*, Volodymyr Kuleshov. "Diffusion Variational Inference: Diffusion Models as Expressive Variational Posteriors". [AAAI](#), 2025 [\[paper\]](#) [\[code\]](#)

[7] **Yingheng Wang**, Shufeng Kong, John M. Gregoire, Carla P. Gomes. "Conformal Crystal Graph Transformer with Robust Encoding of Periodic Invariance". [AAAI](#), 2024 [\[paper\]](#) [\[code\]](#)

[8] **Yingheng Wang**, Yair Schiff, Aaron Gokaslan, Weishen Pan, Fei Wang, Christopher De Sa, Volodymyr Kuleshov. "InfoDiffusion: Representation Learning Using Information Maximizing Diffusion Models". [ICML](#), 2023 [\[paper\]](#) [\[code\]](#)

[9] Yuanqi Du*, **Yingheng Wang***, Yining Huang, Jianan Canal Li, Yanqiao Zhu, Tian Xie, Chenru Duan, John M. Gregoire, Carla P. Gomes. "M²Hub: Unlocking the Potential of Machine Learning for Materials Discovery". [NeurIPS](#), 2023 [\[paper\]](#) [\[code\]](#)

[10] Zhongyu Huang*, **Yingheng Wang***, Chaozhuo Li, Huiguang He. "Going Deeper into Permutation-Sensitive Graph Neural Networks". [ICML](#), 2022 [\[paper\]](#) [\[code\]](#)

[11] **Yingheng Wang**, Yaosen Min, Xin Chen, Ji Wu. "Multi-view Graph Contrastive Representation Learning for Drug-drug Interaction Prediction". [The Web Conf \(WWW\)](#), 2021 [\[paper\]](#) [\[code\]](#) 🏆 [Best Paper Award in AAAI GraphDL 2021](#)

[12] Junjie Yin, Jiahao Dong, **Yingheng Wang**, Christopher De Sa, Volodymyr Kuleshov. "ModuLoRA: Fine-tuning 2-Bit LLMs on Consumer GPUs by Integrating with Modular Quantizers". [ICLR](#), 2025 [\[paper\]](#) [\[code\]](#) 🏆 [TMLR Featured Certificate](#)

[13] He S. Yang*, Weishen Pan*, **Yingheng Wang***, Mark A. Zaydman, Nicholas C. Spies, Zhen Zhao, Theresa A. Guise, Qing H. Meng, Fei Wang. "Generalizability of a Machine Learning Model for Improving Utilization of Parathyroid Hormone-Related Peptide Testing Across Multiple Clinical Centers". [Clinical Chemistry](#), 2023 [\[paper\]](#) [\[code\]](#) 🏆 [The winner solution of the ML challenge \(PTHrP Result Prediction\) at AACC 2022](#)

Others:

[14] Center for AI Safety, Scale AI, HLE Contributors Consortium (incl. **Yingheng Wang**). "A benchmark of expert-level academic questions to assess AI capabilities". [Nature](#), 2026 [\[paper\]](#) [\[code\]](#)

[15] Jiarui Liu, Kaustubh Dhole, **Yingheng Wang**, Haoyang Wen, Sarah Zhang, Haitao Mao, Gaotang Li, Neeraj Varshney, Jingguo Liu, Xiaoman Pan. "Toward Honest Language Models for Deductive Reasoning". [EMNLP](#), 2026 [\[paper\]](#) [Under Review](#)

[16] Taoyuze Lv, Alexander Chen, Fengyu Xie, Jeffrey Meng, **Yingheng Wang**, Bram Hoex, Zhicheng Zhong, Tong Xie. "AtomWorld: A Benchmark For Evaluating Spatial Reasoning In Large Language Models On Crystalline Materials". [ICML](#), 2026 [\[paper\]](#)

[17] Shufeng Kong, Xingru Yang, Yuanyuan Wei, Zijie Wang, Sidie Huang, Hao Tang, Jiuqi Qin, Shuting Lan,

Yingheng Wang, Junwen Bai, Zhuangbin Chen, Zibin Zheng, Caihua Liu, Liang Hao. "MTCMB: A Multi-Task Benchmark for Evaluating LLMs on Knowledge, Reasoning, and Safety in Traditional Chinese Medicine". *Nature npj Digital Medicine*, 2026 [\[paper\]](#) [Under Review](#)

[18] Zhongdi Qu, **Yingheng Wang**, Utku Umur Acikalin, Aaron M. Ferber, Goncalo J. Gouveia, Brandon Bills, Guohui Li, Joshua Kline, Sunandini Yedla, Xiao Wang, Frank C. Schroeder, Carla P. Gomes. "Unsupervised Combinatorial Probabilistic Reasoning: Probabilistic Coin Change Problem". *AAAI*, 2026 [\[paper\]](#)

[19] Shufeng Kong, Zijie Wang, Nuan Cui, Tang Hao, Meng Yiham, Yuan Yuan Wei, Chenfei Fan, **Yingheng Wang**, Zhuo Cai, Yaonan Wang, Yulong Zhang, Yuzheng Li, Zibin Zheng, Caihua Liu, Liang Hao. "MIRNet: Integrating Constrained Graph-Based Reasoning with Pre-training for Diagnostic Medical Imaging". *AAAI*, 2026 [\[paper\]](#)

[20] Christian K. Belardi, Chia-Hao Lee, **Yingheng Wang**, Justin Lovelace, Kilian Q. Weinberger, David A. Muller, Carla P. Gomes. "Improving Electron Ptychography with a Generative Prior". *ICCV*, 2025 [\[paper\]](#)

[21] Yuwei Wan, Yuqi An, Dongzhan Zhou, Jiahao Dong, Chunyu Kit, Wenjie Zhang, Bram Hoex, Tong Xie[†], **Yingheng Wang**[†]. "MatFusion: A Multi-Modal Framework Bridging LLMs and Structural Embeddings for Experimental Materials Property Prediction". *ICLR AI4Mat*, 2025 [\[paper\]](#)

[22] Yuanqi Du*, Arian R Jamasb*, Jeff Guo*, Tianfan Fu, Charles Harris, **Yingheng Wang**, Chenru Duan, Pietro Liò, Philippe Schwaller, Tom L Blundell. "Machine learning-aided generative molecular design". *Nature Machine Intelligence*, 2024 [\[paper\]](#)

[23] Brendan H. Rappazzo, **Yingheng Wang**, Aaron M. Ferber, Carla P. Gomes. "GEM-RAG: Graphical Eigen Memories For Retrieval Augmented Generation". *ICMLA*, 2024 [\[paper\]](#)

[24] Dingshuo Chen, Shuchen Xue, Liuji Chen, **Yingheng Wang**, Qiang Liu, Shu Wu, Zhi-Ming Ma, Liang Wang. "Graffe: Graph Representation Learning Enabled via Diffusion Probabilistic Models". *arXiv*, 2025 [\[paper\]](#)

[25] Zhongyu Huang, **Yingheng Wang**, Chaozhuo Li, Huiguang He. "Growing Like a Tree: Finding Trunks From Graph Skeleton Trees". *IEEE TPAMI*, 2024 [\[paper\]](#) [\[code\]](#)

[26] Tong Zhao, Wei Jin, Yozen Liu, **Yingheng Wang**, Gang Liu, Stephan Gunnemann, Neil Shah, Meng Jiang. "Graph Data Augmentation for Graph Machine Learning: A Survey". *IEEE Data Engineering Bulletin*, 2023 [\[paper\]](#)

[27] Dongsheng Luo, Wei Cheng, **Yingheng Wang**, Dongkuan Xu, Jingchao Ni, Wenchao Yu, Haifeng Chen, Xiang Zhang. "Information-Aware Time Series Meta-Contrastive Learning". *AAAI*, 2023 [\[paper\]](#) [\[code\]](#)

[28] Zhongyu Huang, Changde Du, **Yingheng Wang**, Huiguang He. "Graph-Enhanced Emotion Neural Decoding". *IEEE TMI*, 2023 [\[paper\]](#)

[29] Junwen Bai, Yuanqi Du, **Yingheng Wang**, Shufeng Kong, John M. Gregoire, Carla P. Gomes. "Xtal2DoS: Attention-based Crystal to Sequence Learning for Density of States Prediction". *NeurIPS AI4Science*, 2022 [\[paper\]](#) [\[code\]](#)

[30] Zhongyu Huang, Changde Du, **Yingheng Wang**, Huiguang He. "Graph Emotion Decoding from Visually Evoked Neural Responses". *MICCAI*, 2022 [\[paper\]](#)

[31] **Yingheng Wang**, Yaosen Min, Erzhuo Shao, Ji Wu. "Molecular Graph Contrastive Learning with Parameterized Explainable Augmentations". *BIBM*, 2021 [\[paper\]](#)

[32] **Yingheng Wang**, Xin Chen, Yaosen Min, Ji Wu. "MolCloze: A Unified Self-supervised Molecular Structure Pre-training Model for Chemical Property Prediction". *BIBM*, 2021 [\[paper\]](#)

[33] Erzhuo Shao*, Jie Feng*, **Yingheng Wang**, Tong Xia, Yong Li. "One-shot Transfer Learning for Population Mapping". *CIKM*, 2021 [\[paper\]](#)

[34] S. Kong, T. Yu, Y. Wei, C. Liu, J. Bai, **Yingheng Wang**, M. Grimson, D. Fink, C. P. Gomes. "STELLAR: Spatio-Temporal Environmental Learning with Latent Alignment and Refinement for Long-Tailed Species

Distribution Modeling". [IJCAI](#), 2026

[35] S. Kong, F. Deng, X. Dong, C. Liu, W. Chen, **Yingheng Wang**, D. Cao, A. Oliva, A. Fernandez-Ruiz, C. P. Gomes. "Machine Learning Methods for Studying Latent Neural Activity Dynamics". [IJCAI](#), 2026

[36] Chia-Hao Lee, Christian K. Belardi, **Yingheng Wang**, Justin Lovelace, Kilian Q. Weinberger, David A. Muller, Carla P. Gomes. "3D Ptychographic Inverse Imaging with Generative Diffusion Models". [Microscopy and Microanalysis](#), 2025 [\[paper\]](#)

[37] Y. Min, Y. Wei, P. Wang, N. Wu, S. Bauer, S. Zheng, Y. Shi, **Yingheng Wang**, X. Wang, D. Zhao, J. Wu, J. Zeng. "From Static to Dynamic Structures: Improving Binding Affinity Prediction with Graph-Based Deep Learning". [Advanced Science](#), 2024 [\[paper\]](#)

[38] B. Yang, T. Meng, X. Wang, J. Li, S. Zhao, **Yingheng Wang**, S. Yi, Y. Zhou, Y. Zhang, L. Li, L. Guo. "CAT Bridge: An Efficient Toolkit for Compound-Transcript Association Mining from Multi-Omics Data". [Giga-Science](#), 2024 [\[paper\]](#)

[39] C. Li, T. He, **Yingheng Wang**, et al.. "Pipeline Image Haze Removal System Using Dark Channel Prior on Cloud Processing Platform". [International Journal of Computational Science and Engineering](#), 2020 [\[paper\]](#)

[40] T. Teng, **Yingheng Wang**, et al.. "Mathematical Modeling and Simulation on the Stimulation Interactions in Coalbed Methane Thermal Recovery". [Processes](#), 2019 [\[paper\]](#)

Awards & Honors

• ICML Gold Reviewer	2026
• Outstanding Graduate Teaching Assistant	2026
• TMLR Featured Certificate	2025
• Google AI Academic Program GCP Credit Award	2024
• TMLR Featured Certificate	2024
• The Winner of the ML Challenge (PTHrP Result Prediction), AACC	2022
• The Participation Scholarship, ICML	2023, 2022
• Presidential Life Science Fellowship, Cornell University	2022
• Best Paper Award, AAAI DLG Workshop	2021
• Graduate Scholarship of Tsinghua University	2021
• Outstanding Graduate Award (sole recipient)	2020
• Best Undergraduate Thesis Award (sole recipient)	2020
• Outstanding Undergraduate Researcher Award (10 recipients)	2019
• Meritorious Winner, in Mathematical Contest In Modeling (MCM)	2019
• National Scholarship	2018
• First-Class Academic Scholarship	2017

Work Experience

Grand Challenge, Amazon	NYC, NY, 2024
<i>Applied Scientist Intern</i>	<i>policy optimization for antibody sequence design</i>

- Managers: [Marcus D. Collins](#), [Shang Shang](#)
- Mentors: [Tommi S. Jaakkola](#), [Karla-Luise Herpoldt](#), [Can Chen](#)

AWS AI Research & Education, Amazon	NYC, NY, 2024
<i>Applied Scientist Intern</i>	<i>long-context modeling of protein-protein interactions</i>

- Managers: [Zichen Wang](#), [Huzefa Rangwala](#)

- Mentors: [George Karypis](#), [Alessandro Achille](#), [Luca Zancato](#)

NEC Laboratories America

Research Intern

Princeton, NJ, 2021
times-series contrastive learning

- Manager: [Wei Cheng](#)
- Mentor: [Dongsheng Luo](#)

University of Washington

Research Assistant

Seattle, WA, 2021
drug relation mining with multi-modal data

- Advisor: [Sheng Wang](#)

Johns Hopkins University

Research Assistant

Baltimore, MD, 2021-2022
GNNs for brain network analysis

- Advisor: [Joshua T. Vogelstein](#)
- Mentor: [Sambit Panda](#)

Microsoft Research Asia

Research Assistant

Beijing, 2020-2022
expressivity and long-range capability of GNNs

- Manager: [Chaozhuo Li](#)

Tsinghua University

Research Assistant

Beijing, 2020-2022
machine learning for drug discovery

- Advisor: [Ji Wu](#)

Professional Service

Area Chair: NeurIPS (AI4Science), ICML (AI4Science)

Conference Reviewer: ICML, NeurIPS, ICLR, AAAI, AISTATS, UAI, CVPR, ICCV, ECCV, COLM, KDD

Journal Reviewer: IEEE Transactions on Knowledge and Data Engineering (TKDE), Transactions on Machine Learning Research (TMLR), Pattern Recognition, Bioinformatics, Journal of Biomedical Informatics

Workshop Reviewer/PC: NeurIPS (Math-AI, AI4Science, GenBio, GLFrontiers), ICLR (AGI, AI4Science), ICML (AI4Science), KDD (Health Day)

Others: Co-organizer of Learning on Graph Conference (LoG 2025, 2024, 2023) local meetup, Cornell CS PhD Admission Committee Member (2025, 2024), Cornell CS Visit Day CZar

Invited Talk

- **Grand Challenge @ Amazon**, *AbMPO*, hosted by Marcus D. Collins 2025
- **Machine Learning Day @ ASU**, *LC-PLM*, hosted by Yixuan He 2025
- **Multi-omics Reading Group @ MILA**, *LC-PLM*, hosted by Xinyu Yuan 2025
- **AI4Protein Seminar**, *LC-PLM*, hosted by Jun Li 2024
- **AWS AI Research & Education @ Amazon**, *LC-PLM*, hosted by Zichen Wang 2024
- **Toyota Research Institute**, *CrystalFormer & Xtal2DoS*, hosted by Santosh Suram 2023
- **Weill Cornell Medicine**, *InfoDiffusion*, hosted by Fei Wang 2023
- **Florida International University**, *InfoDiffusion*, hosted by Dongsheng Luo 2023
- **American Association for Clinical Chemistry**, *Championship Solution for the PTHrP Result Prediction Challenge*, hosted by Nicolas Spies 2022
- **AAAI GraphDL 2021**, *MIRACLE*, hosted by Lingfei Wu 2021

Teaching

- **CS 2770: Computational Sustainability**, Cornell University 2026
Undergraduate Level, *Teaching Assistant*, 🏆 Outstanding Graduate TA Award
- **CS 6703: AI for Science**, Cornell University 2023
Graduate Level, *Teaching Assistant*
- **Data Structures and Algorithms**, Tsinghua University 2021
Undergraduate Level, *Teaching Assistant*
- **Introduction to Database**, Tongji University 2019
Undergraduate level, *Teaching Assistant*

Mentorship

- Jiarui Liu** 2025
- MS at CMU (currently Ph.D. student at CMU LTI).
 - Mentored Jiarui on RL-finetuning LLMs for deductive reasoning.
- Junjie (Oscar) Yin** 2023-2024
- Undergrad at Johns Hopkins (currently Ph.D. student at UW).
 - Mentored Oscar on finetuning quantized LLMs.
- Jiahao Dong** 2023-2024
- MS at Cornell Tech (currently ML Engineer at Distyl AI).
 - Mentored Jiahao on finetuning quantized LLMs.
- Jianan (Canal) Li** 2023
- Undergrad at Cornell (currently Ph.D. student at Berkeley).
 - Mentored Canal on benchmarking geometric learning methods and datasets for materials.
- Yining Huang** 2023
- Undergrad at Northwestern (currently Ph.D. student at UPenn).
 - Mentored Yining on benchmarking geometric learning methods and datasets for materials.
- Bowen Yang** 2022-2023
- MS at Georgetown (currently Ph.D. student at UAlberta).
 - Mentored Bowen on drug relation extraction using multi-modal data.
- Zhengchao Luo** 2021-2022
- Undergrad at HUST (currently Ph.D. student at PKU).
 - Mentored Zhengchao on predicting quantum chemical properties of molecules using GNNs.